Code No: 181AP **R22**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, January/February - 2024
PROGRAMMING FOR PROBLEM SOLVING
(Common to CSE(AI&ML), CSE(IOT), AI&DS, AI&ML)

Time: 3 Hours **Max. Marks: 60**

Note: This question paper contains two parts A and B.
i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

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PART- A

(10 Marks)

1.a)	List bitwise operators in C.	[1]
b)	What is the purpose of type conversion?	[1]
c)	What is an array?	[1]
d)	What is the significance of the **strcpy** function?	[1]
e)	What is the purpose of the preprocessor in C?	[1]
f)	How can you check if a file exists before opening it in C?	[1]
g)	What is a function prototype?	[1]
h)	Define the term "function signature".	[1]
i)	What is the time complexity of bubble sort?	[1]
j)	What is the main advantage of binary search over linear search?	[1]

PART - B

(50 Marks)

2.a)	Write an algorithm for finding the roots of a quadratic equation.	
b)	Discuss the importance of flowchart and pseudocode in designing algorithm.	[5+5]
	OR	
3.a)	Describe the different storage classes available in C.	
b)	Explain various types of errors with examples.	[5+5]
4.a)	Explain how to define and use an enumeration data type in C.	
b)	Compare character arrays and string literals. How are strings handled as arrays of characters in C?	[4+6]
	OR	
5.a)	Describe the process of creating and initializing a one-dimensional array. Give an example.	
b)	Discuss the use of pointers in self-referential structures.	[6+4]

6.a) Write a C program to check the end-of-file condition using the **feof** function.
b) Compare and contrast the **#ifdef** and **#ifndef** directives in C. [5+5]

OR

7.a) Differentiate between text and binary file modes in C.
b) Write a program to append data to an existing text file in C. [5+5]

8. Describe memory allocation and de-allocation using **malloc()** and **free()** functions. Write suitable example program. [10]

OR

9.a) Explain call by reference mechanism with an example.
b) Write a C program to calculate the factorial of a given number using recursion. [5+5]

10. Explain the binary search algorithm in detail. Provide an example. [10]

OR

11. Explain the key steps involved in the Insertion sort algorithm with an example. [10]

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